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Wheeled Mobile Robot Modeling for Local Navigation Using System Identification

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***Abstract*—In this work we investigate wheeled mobile robot (WMR) kinodynamic modeling using system identification techniques. The larger context is the development of a novel local navigation algorithm based on long kinodynamic trajectories which account for vehicle drivability. A high-fidelity physics-based differential-drive robot model is used to generate command velocity and ground truth data sets, which are subsequently analyzed using multiple system identification models. Parametric linear ARX and ARMAX models are considered. Experimental results using multiple training and validation data sets corresponding to different driving scenarios and vehicle loading are explored. The resulting model-generated WMR paths demonstrate excellent performance.**

***Index Terms*—System identification, modeling, wheeled mobile robot, local navigation.**

I. Introduction

Wheeled Mobile Robots (WMR) are an important class of autonomous vehicle, which currently finds application in a broad array of contexts ranging from Search and Rescue (SAR), military exploration and mapping, warehouse stock movement, and agricultural planting and harvesting. A central element of WMR motion control is the navigation system which typically comprises a global and a local navigator. The local navigator generates an immediately-responsive behavior to the evolving environment, while the global navigator establishes a suitable longer-term path to the next waypoint of interest. The latter is generally computationally intensive, but is run less frequently or in the background. On the other hand, the former needs to be fast, since it addresses the current situation and is continuously active [1].

In order to generate an accurate motion trajectory for a given set of inputs, a high quality vehicular model must be available. Such models are commonly developed via detailed mechanical and electrical physics-based description of each of the robot's components and their interactions [2]. The result is a complex set of linear and nonlinear differential equations, which are solved numerically.

Such high-fidelity vector differential-equation-based models are computationally intensive and thus are not generally suitable for real-time use. Rather, they more often serve as a reference result for vehicular design and development [3]. This presents a challenge for real-time robot motion planning which can be addressed in a number of ways. One common approach is to use the robot's basic kinematic model and assume that desired velocities are achieved instantaneously. This results in simple constant-curvature paths which, since they do not account for Newtonian mechanics, can only be used with short-interval algorithms which "refresh" the input commands at a high rate to manage the attendant trajectory errors. Alternatively, if complete or more "dynamically aware" model solutions are used, longer and more accurate trajectories can be achieved.

In this work then, we explore the use of System Identification (SI) in combination with a physics-based high-fidelity reference robot model to develop a parametric approximation (ARX & ARMAX) of the reference model. The robot dynamics are shown to be well represented in the forward motion trajectories generated for linear and angular velocity input commands. Motion paths generated with the SI vehicle model are compared with those produced by the reference model to demonstrate the efficacy of the method.

The paper is organized as follows. Section II summarizes background work via a brief listing of the most relevant literature. Section III presents the reference robot simulation environment, a summary of the kinematic motion equations, and a discussion of the effects of dynamics consideration on motion. It also outlines the structure of the parametric models evaluated. This is followed by Section IV in which we present experimental results that show the effects of inclusion of vehicular dynamics on shaping of path trajectories and demonstrate how well the SI-WMR models track the physics-based reference robot characteristics. Finally, Section V presents conclusions and summarizes ongoing work.

II. Background Research

In this section the principal work related to our efforts is briefly discussed. There are two main topic areas to consider: local navigation for WMRs and system identification.

A. Local Navigation

The SI modeling and trajectory generation work presented in this paper is associated with a novel local navigator first introduced in [1]. Related work includes the Dynamic Window

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Approach (DWA) [4], Driving With Tentacles (DWT) [5], and the Trajectory Space (TS) work presented in [6]. Also, there are connections with a series of publications concerned with the construction of a state lattice comprising a small set of motion primitives or trajectories, which can be concatenated as part of a motion-predictive-control strategy [7]–[10]. These latter approaches seek to solve the *inverse* kinematic or kinodynamic motion equations (or a parameterized representation), so as to generate control inputs to drive a robot on various short trajectories, each of which includes boundary constraints. Recent efforts explore novel adaptive strategies to reduce the associated computation [11].

A review of the literature on robot motion planning and navigation, reveals few comprehensive efforts formally accounting for vehicle dynamics effects in combination with trajectory-based reactive local navigation algorithms [12], [13]. Our SI-based local navigation approach projects trajectories in position rather than velocity space, and does not make simplifying assumptions about their generation, producing more realistic estimates of probable vehicle paths. This rich set of kinodynamically feasible forward-trajectory candidates more accurately predicts vehicle behavior at higher speeds, where dynamic responses are more prevalent, permitting longer intervals of deliberative driving (with concomitant savings in computational resources).

B. System Identification

System Identification (SI) is a powerful and increasingly popular approach for estimating the behavioral characteristics of dynamical systems using available input and output data sets. In particular, linear and non-linear systems may be parameterized to facilitate reduced order modeling of complex systems [3]. Due to the complexity of physical vehicles, SI is commonly considered when developing algorithms for the control, modeling, and trajectory development of robot systems [14]–[24]. A systems identification approach is useful in quickly providing reliable and sufficiently accurate models, without a deep knowledge of the underlying physics and mathematical representations.

In this work, we explore the use of Autoregressive and Autoregressive Moving Average with eXogenous input models (ARX, ARMAX) [25, pp. 71] for the estimation of relevant parameters for differential-drive wheeled mobile robots. The goal is to develop SI-derived models which can be used to create trajectories in response to linear and angular velocity commands, which are sufficiently accurate to permit ballistic driving for longer intervals. These forward-trajectories provide short-range smooth paths for local navigation.

III. Robot Simulation and Modeling

The algorithms in this work were tested on a high-fidelity Webots® model of the Pioneer P3-DX®robot [26]. Cyberbotics maintains Webots as its main product [27]. For simulating rigid body dynamics and collision detection, Webots® uses the ODE (Open Dynamics Engine) library which enables the simulator to accurately represent the robot's physical properties.

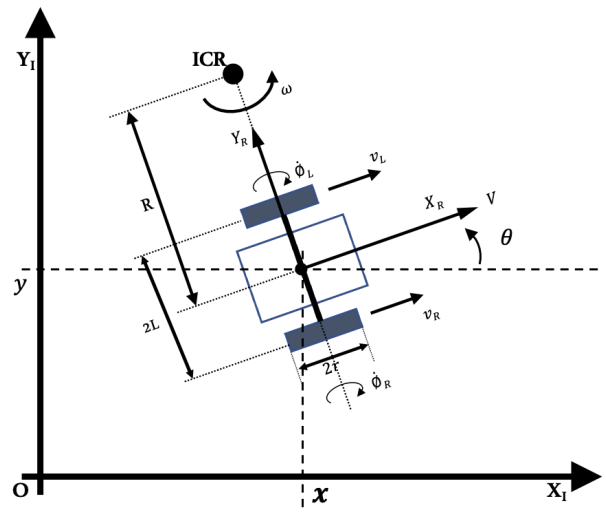

Fig. 1. Differentially-steered robot

A. Kinodynamic Motion Modeling

The kinematic equations of motion for a differentially steered robot [28, p. 57] [29, p. 43] are summarized in this section. The variables of interest are shown in Figure 1.

The robot's forward velocity is the average of the velocities of the center of the right and left wheels, which themselves are a function of the rotational velocities of the two wheels, as shown in (1). The latter are directly driven by the two drive motors of the vehicle.

$$v = \frac{v_R + v_L}{2} = \frac{r\dot{\varphi}_R + r\dot{\varphi}_L}{2} \tag{1}$$

The robot's rotational velocity depends on the difference between the right and wheel velocities scaled by the spacing between the drive wheels, as shown in (2).

$$\dot{\theta} = \omega = \frac{v_R - v_L}{2L} = \frac{r\dot{\varphi}_R - r\dot{\varphi}_L}{2L} \tag{2}$$

The robot will trace a circular path at any instant, whose radius is given by (3), based on the forward and turn velocities at that instant:

$$R = \frac{v}{\omega}. \tag{3}$$

Its instantaneous velocities with respect to the global frame can then be obtained by standard projections as shown in (4), (the robot's kinodynamic equations):

$$\begin{bmatrix} \dot{x_I} \\ \dot{y_I} \\ \dot{\theta_I} \end{bmatrix} = \begin{bmatrix} cos\,(\theta) & 0 & 0 \\ sin\,(\theta) & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ 0 \\ \omega \end{bmatrix} \tag{4}$$

If the robot's velocities could be instantaneously changed, the instantaneous curvature of its circular path could be altered on demand (as in 3), thus enabling kinematic trajectories to be stitched together, as desired, without any transition phase.

However, complications arise when one considers that it is only possible to change velocities instantaneously if either the available torques from the motor are very large, the mass of the robot is negligible, or both. That is, when Newton's Law is

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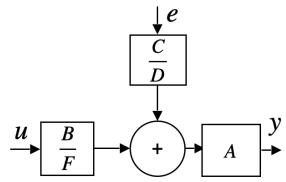
applied in practice, there will inevitably be a transitional delay between two (v, ω) kinematic states, as the motors attempt to move the robot from the old velocity state to the new velocity state using finite accelerations. The inevitability of what might be called this "lazy phase" is enough to make the task of trajectory planning significantly more complex, particularly when considered in conjunction with additional practical aspects, as discussed below.

Assume that the robot needs to transition from its current kinematic state to a target kinematic state. This will map to corresponding current and target right and left wheel velocities through an inversion of (1) & (2). It can be stated that in general the change in kinematic state will require different amounts of change in right and left wheel velocities. Also, that the corresponding transition times will be different, since they are a function of the available accelerations of the two motors, which are dependent on their individual load torques as well as rotational speeds. This process introduces non-linearity into the cause-effect linkage, which is further exacerbated by limits on the available acceleration. A key aspect of this is that error in turn angle estimation causes path deviation to build up! The worst case manifestation of this is when the robot is making a hard left turn at a high speed and is then asked to switch to a hard right turn, or the reverse. There will not be an instantaneous transition from one curvature to the other.

B. Parametric Modeling

For robot model parameter estimation, some of the most popular and effective models to consider are the linear ARX, ARMAX, and Box-Jenkins models [14], [16], [18]–[20]. To estimate the coefficients of a parametric WMR model, one must generate a rich input/output data set, select model configurations (regressors, delays, offsets, etc.), and estimate the parameters using an identification algorithm, e.g. a method from the family of prediction-error identification techniques (PEM) [25, p. 171].

Central to this process is the careful selection of system inputs, so as to ensure that the various system modes are sufficiently excited. That is, one must select input commands (e.g. uniformly distributed random binary, step and multi-step sequences with random height and duration, chirp signals, etc.), that are capable of triggering full-spectrum vehicle responses. Subsequently, the vehicle's resulting path would be recorded, and the input/output data sets would then be processed with commercial System Identification (SI) algorithms to generate an approximate system model. For example, the Matlab® System Identification Toolbox (by Mathworks®) can estimate black-box model parameters for Multiple-Input Multiple-Output (MIMO) systems using input/output data sets. The data-driven vehicle model estimate is then used to generate the desired family of trajectories, something that would not be feasible with actual vehicle runs, because of the large number required. The estimated model produces simulated vehicle reactions in response to the input command vectors. Hence, using this information, the resultant trajectories can be generated based on the robot's current state (v_i, ω_i) and the



Common Black-box Models

Polynomials Used	Model Structure Name
B	FIR
AB	ARX
ABC	ARMAX
AC	ARMA
BFCD	Box-Jenkins

Fig. 2. General Linear Model structures and Configurations [25, pp. 77]

target forward and rotational velocities (v_t, ω_t).

The most complete parameterization of a general output-error parametric model can be written as:

$$A(q)y(k) = \frac{B(q)}{F(q)}u(k) + \frac{C(q)}{D(q)}e(k) \tag{5}$$

where $y(k)$ is the output, $u(k)$ is the input, and $e(k)$ is disturbance. Figure 2 presents a data flow block diagram corresponding to (5), with the most common named-transfer-function configurations. To illustrate, only $A(q)$ and $B(q)$ are present for the ARX model yielding regressions on $y(k)$ and $u(k)$ respectively (q is a delay operator) [25, p. 19]. For this work, the ARX, ARMAX, and Box-Jenkins model structures were evaluated over a large range of data sets. Ultimately, the ARX structure was the most efficient in terms of modeling accuracy and parameter number. The ARMAX and Box-Jenkins configurations provided only modest improvement by permitting further filtering of the disturbance input $e(k)$. For WMR ARX modeling, there is an input command vector $(v_{in}, \omega_{in})^T$ assigned as $\mathbf{u}(k)$, as well as two output signals $(v_{out}, \omega_{out})^T$ which comprise the output vector $\mathbf{y}(k)$. In this Multiple Input Multiple Output (MIMO) case then, the polynomials $A(q)$ & $B(q)$ become 2×2 coefficient matrices $\mathbf{A_i}$ & $\mathbf{B_i}$. Thus the MIMO ARX model may be written as:

$$\mathbf{y}(k) = \sum_{i=1}^{n} \mathbf{A_i}\mathbf{y}(k-i) + \sum_{i=1}^{m} \mathbf{B_i}\mathbf{u}(k-i) + \mathbf{e}(k) \tag{6}$$

where $\mathbf{e}(k)$ is a 2×1 vector disturbance sequence.

IV. Experimental Study

A. Estimation and Modeling Results

A Webots® Pioneer P3-DX® robot model was used to generate a range of input/output data sets for this study. In particular, random-height multi-step sequences were generated as $(v_{in}, \omega_{in})^T$ command inputs. Sequences ranged from 4000 to 10000 points with a sampling interval of 0.01 seconds. Step response and Bode response curves for the Pioneer P3-DX® model were used to identify a parsimonious sampling interval for the simulation. Also, one and two-second multi-step intervals were selected to permit the vehicle motion to stabilize after each input command. Model quality was judged by prediction-error sums for validation-data sets, as well as driving performance compared to that of the Webots® model, for a variety of complex paths and simple trajectories.

The SI models were trained with a ground-truth output data set resulting from the application of an input sequence comprising fifty 2-second duration, multi-step $(v_{in}, \omega_{in})^T$ inputs.

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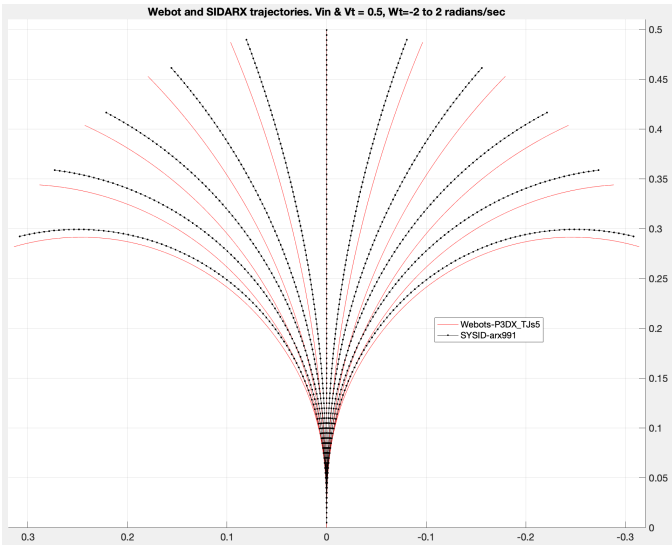

Fig. 3. 3-second trajectories for $\omega_{in} = 0$ and fixed v_{in} with varying ω_{final}

P3-DX input-sequence→motion simulations were generated (using a 0.01 second resolution). SI models estimated from training sequences were then tested with separate validation sequences. ARX, ARMAX, and Box-Jenkins model structures were explored, and as there was no notable improvement (less than 0.4%) in the results for the latter models, only the ARX results are discussed going forward. For the base robot, the model fit (based on prediction-error identification) ranged from 85.1% to 94.76% for v and 67.2% to 95.68% for ω depending on the validation data set used.

Additional SI models were similarly trained and tested for a Webots®-based P3-DX® configured with a 10kg additional mass (the unloaded P3-DX robot weighs 9kg). For the 10kg loaded robot, the fit percentages ranged from 78.4% to 92.15% for v and 44% to 85.72% for ω.

The best ARX model for the unloaded robot had 9th order regressions on $(\mathbf{y_{11}})$, $(\mathbf{y_{22}})$, $(\mathbf{u_{11}})$ and $(\mathbf{u_{22}})$ with 3rd order regressions for $(\mathbf{y_{12}})$ & $(\mathbf{y_{21}})$ and 1st order regressions for $(\mathbf{u_{12}})$ & $(\mathbf{u_{21}})$. These modeling parameters were selected while monitoring validation-data-set fit quality to avoid over fitting. Also, the 1-step-prediction-residual cross correlation (between residuals and inputs) was monitored, so as to ensure that correlations were within the 99% confidence intervals. The best ARX model for the 10kg loaded robot had 15th order regressions on $(\mathbf{y_{11}})$, $(\mathbf{y_{22}})$, $(\mathbf{u_{11}})$ and $(\mathbf{u_{22}})$ with 3rd order regressions for $(\mathbf{y_{12}})$, $(\mathbf{y_{21}})$, $(\mathbf{u_{12}})$ and $(\mathbf{u_{21}})$.

The trajectory plot in Figure 3 demonstrates how closely the simulated vehicle response tracks the Webots®-based P3-DX® model for a range of target ω_t values for $\omega_{in} = 0$. Worst-case long-path comparisons for unloaded and 10Kg loaded models are provided in Figure 4 which displays ideal (kinematic), Webot®(ground-truth), and SI-model paths (starting from the bottom & ending at the top). These paths were generated by training the model with a moderate 2-second command-interval-sequence and then simulating with an aggressive 1-second command-interval-sequence comprising maximum swings in both v_{in} and ω_{in} from one second to the next – such rapid input changes cause greater robot motion transients and do not permit the vehicle to reach steady state. The curves represent 50 seconds of ballistic (no feedback) driving and approximately 18 meters of travel. When moderate, 2-second command sequences are used for validation, it is common to obtain sub-centimeter accuracy for 1 second of travel with less than 5cm path deviation for 5 seconds. It is typical in the literature for 10ms path-refresh intervals to be used [30] to update model-predictive-control trajectories, so a one-second trajectory update represents a significant refresh-rate reduction.

V. Conclusions and Future Work

In this work we investigated wheeled mobile robot (WMR) kinodynamic modeling using system identification in order to support the parallel development of a novel local navigation algorithm based on the use of relatively long trajectories. A high-fidelity robot model was effectively used to generate command velocity and ground truth data sets which were used to identify linear ARX models. The experimental study demonstrated highly effective modeling results and presented WMR forward trajectory and path tracking performance data. The proposed approach is very useful for work wherein robot platforms must be used without the benefit of detailed physics models and where computational efficiency is necessary.

Future work will implement the proposed methodology by collecting data with an actual physical robot driven with short command sequences. The resulting SI models will then be implemented with a trajectory-based local navigator to quantify performance in realistic environments. Additionally, nonlinear models will be investigated to determine whether they can provide any additional benefit.

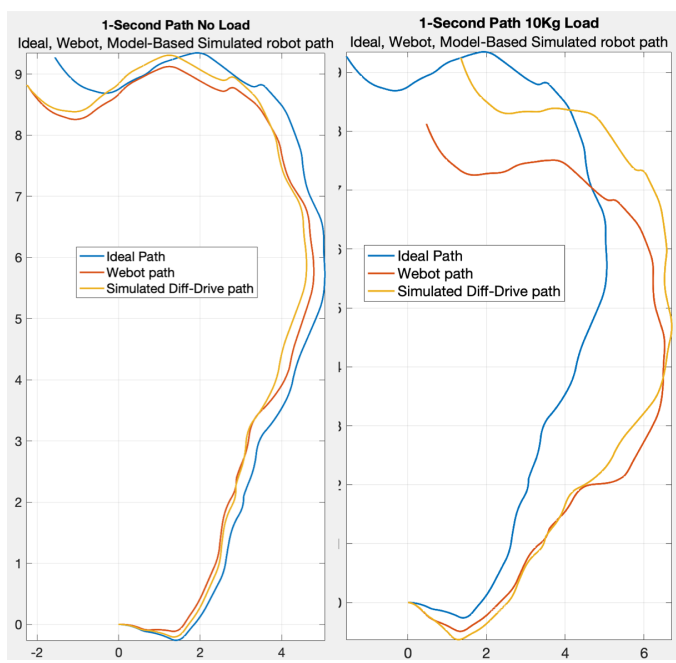

Fig. 4. 50-Second path (meters) unloaded & loaded ARX Model (ideal=kinematic)

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